Cardiovascular Disease Dataset

Exploratory data analysis: findings report | 68,666 records, 13 columns, target: *cardio*

Key takeaways

- The data is nearly balanced between the two outcome classes, and its quality issues are minor (33 missing cells, 24 repeated rows).
- **Blood pressure** and **age** separate the classes most clearly; **weight** shows a modest shift; **height** shows almost none.
- **Cholesterol** and **glucose** levels are strongly related to disease rate; gender, smoking and alcohol are not.
- All numeric features contain possible outliers. They were kept, pending a plausibility review.

1. Data quality: missing values, repeats and recording issues

The audit found **33 missing values** in total, **27 in height** and **6 in weight**, together with **24 rows** that repeat an earlier identical row. No column is constant or completely empty. These points were noted for review, but the repeated rows were **kept**: two records can share exactly the same attributes without either one being an error. The missing values and the way the data was recorded should be kept in mind when reading the results below.

2. Class balance

The two outcome classes are close to equal in size. The larger class holds 50.53% of the rows and the other 49.47%. Predicting the larger class for every row would therefore be right 50.53% of the time. That figure is a reference point only, not the result of a validated model.

Class	Rows	Share
No cardiovascular disease	34,697	50.53%
Cardiovascular disease	33,969	49.47%

3. Shape of the numeric features

The five numeric features differ in range and shape. Age is concentrated in middle and later adulthood. Height and weight each have their own pattern and carry the few missing values mentioned above, while the blood pressure readings span a broad set of recorded values. Histograms help locate where values concentrate and whether gaps or spikes exist, but on their own they say nothing about how the disease classes differ.

4. Frequency of categories

Category sizes are uneven. Gender 1 appears more often than gender 2; cholesterol level 1 and glucose level 1 dominate their variables; smoking and alcohol use are comparatively rare; and most people are recorded as physically active (active = 1). Such differences in group size should be remembered when comparing disease rates between categories.

5. Unusual values

Box plots reveal values beyond the whiskers in all five numeric features: unusually low and high heights and weights, extreme systolic (ap_hi) and diastolic (ap_lo) pressures, and a handful of low ages. Nothing was removed at this stage, since an unusual value is not necessarily a mistake. Whether each is plausible, and how much it matters for classification, should be checked before any removal.

Feature	Lower fence	Upper fence	Flagged rows	Flagged %
age_years	33.15	73.55	4	0.006
height	142.5	186.5	478	0.696
weight	39.5	107.5	1,739	2.533
ap_hi	90	170	1,005	1.464
ap_lo	65	105	3,499	5.096

6. Numeric features compared across classes

6.1 Where each class sits along the range

- **age_years**: a clear shift. Younger ages (for example under 48) are mostly people without disease, whereas disease cases grow noticeably more common between roughly 55 and 65 years.
- **ap_hi and ap_lo**: strong separation. People without disease cluster near typical values (ap_hi about 120, ap_lo about 80); people with disease move clearly toward higher pressures.
- **weight**: the classes overlap heavily, with the disease group leaning slightly toward higher weights.
- **height**: the two distributions are almost the same, so height alone does not separate the classes.

6.2 Medians, spread and outliers

- **age_years**: median age is about 56 in the disease group versus about 52 in the other group.
- **ap_hi and ap_lo**: the disease group has noticeably higher medians and wider interquartile ranges, and both groups contain many flagged outliers.
- **weight**: medians overlap strongly; the disease group is slightly higher and has more extreme heavy values.
- **height**: medians, spreads and whiskers are nearly identical, so height has little value for telling the classes apart.

6.3 Averages and extremes

Feature	Mean (no disease vs disease)	Min	Max
age_years	about 51 vs about 55	about 29 vs about 39	about 65 in both
ap_hi / ap_lo	higher in the disease group	ap_hi minimum higher with disease	about 240 / about 150 in both
weight	about 71 kg vs about 76 kg	30 kg in both	200 kg in both
height	about 164 cm in both	100 cm in both	about 200 cm in both

7. Categorical features compared across classes

Looking at the share of each class within every category:

- **Cholesterol and glucose** are strongly informative. The disease share climbs steeply with cholesterol and reaches nearly 76% at level 3.
- **Gender, smoking and alcohol** stay near 50% in every category, so each has little discriminating power by itself.
- **Physical activity**: inactive people (active = 0) show a slightly higher disease rate, about 53%, than active people.

Feature / category	Rows	No disease	Disease
Cholesterol 1	51,497	56.46%	43.54%
Cholesterol 2	9,300	40.37%	59.63%
Cholesterol 3	7,869	23.74%	76.26%
Glucose 1	58,386	52.44%	47.56%
Glucose 2	5,065	41.15%	58.85%
Glucose 3	5,215	38.22%	61.78%
Active = 0	13,502	46.75%	53.25%
Active = 1	55,164	51.46%	48.54%

Percentages are computed within each category (each row sums to 100%).